

Forthcoming Short-term Certificate Courses on

PCB Design, Robotics, Basics of Electronics & Microcontrollers

At Bangalore & New Delhi
EFY Tech Centers

Who can benefit?

Electronics engineering students, hobbyists, working professionals wanting to learn more and job seekers

What do they get out of it?

Job seekers and working professionals add to their qualifications, and students and hobbyists assemble exciting engineering projects on their own



Courses duration up to 10 days



First come first served



Limited seats



Don't wait, join now

PCB Design (Delhi & Bangalore)

Duration: 5 Days (30 hours); **Schedule:** Delhi: 1 Sept - 5 Sept / 12 Oct - 16 Oct / 26 Oct - 30 Oct / 2 Nov - 6 Nov / 16 Nov - 20 Nov; **B'lore:** 28 Sept - 3 Oct
Course Fee: Rs 4000/-* **Eligibility:** Engineering students, working engineers, working diploma holders, hobbyists

COURSE DESCRIPTION:

- Learn to use EAGLE PCB design software, by gaining understanding of the basic concepts of PCB design and skills in drawing the schematic, editing and routing and creation of a library • Make a PCB layout of an electronic circuit with various components such as a microcontroller, ICs, transistors, diodes, passive components, switches and connectors

Certificate awarded by EFY

Microcontroller: Microchip PIC16F877A (Delhi & Bangalore)

Duration: 10 days (60 hours); **Schedule:** Delhi: 7 Sept - 18 Sept / 5 Oct - 16 Oct / 19 Oct - 29 Oct / 2 Nov - 12 Nov; **B'lore:** 14 Sept - 24 Sept / 9 Nov - 20 Nov; **Course Fee:** Rs 6000/-* **Eligibility:** Engineering students, working engineers, working diploma holders

COURSE DESCRIPTION: • Understand the architecture of PIC16F877A microcontroller • Understand the memory organisation, special-function registers and on-chip peripherals • Learn instruction set and writing of codes in Assembly language • Learn the simulation, real-time debugging and programming processes with the help of MPLAB: ICD 2 and PIC Development board • Learn integration of common peripherals with microcontroller through hands-on lab exercises

Certificate awarded by Microchip Inc., USA

Accommodation can be arranged in a private hostel/guest house on prior request.

Mode of payment: Please add 10.3% Service Tax while paying fee. The fee may be paid in cash or through a DD in the name of **EFY Enterprises Pvt Ltd**, payable at Delhi or Bangalore. You can pay the entire fee at one go or register by paying Rs 1000 as advance towards the course fee.

Payments should be sent to:

Ms Jayashree, EFY TECH CENTER

No. 9, 17 Main First Cross, HAL Second Stage, Indira Nagar, Bangalore. Phone: 080-25260394, 25260023



Ms Rinku, EFY TECH CENTER

D88/5 Okhla Industrial Area, Phase 1, New Delhi 110020

E-mail: eftytech@efyindia.com Phone: 011-26810601 to -603

Basics of Electronics (Delhi & Bangalore)

Duration: 5 days (30 hours); **Schedule:** Delhi: 1 Sept - 5 Sept / 5 Oct - 9 Oct / 26 Oct - 30 Oct / 2 Nov - 6 Nov / 23 Nov - 27 Nov; **B'lore:** 7 Sept - 11 Sept / 26 Oct - 30 Oct / 23 Nov - 27 Nov; **Course Fee:** Rs 4000/-* **Eligibility:** Engineering students, diploma holders, 12th pass with physics, hobbyists

COURSE DESCRIPTION:

- Know about electronic components (both active as well as passive)
- Learn important circuit blocks which will help you in developing your own minor project • Familiarise with basic test equipment for testing and fault-finding • Learn soldering through assembly of an EFY kit

Certificate awarded by EFY

Renesas R8C Microcontroller (Bangalore)

Duration: 10 days (60 hours); **Schedule:** Bangalore: 12 Oct - 22 October; **Course Fee:** Rs 6000/-* **Eligibility:** Done a basic course in C language programming

COURSE DESCRIPTION:

Architecture of R8C microcontroller; C language programming for integration of peripherals; Hands-on training on use of Renesas Starter Kit, board, E8 debugging emulator and application board in HEW environment (HEW integrates various tools such as compiler, assembler, debugger and editor into a common GUI). Learn to evaluate Renesas microcontroller performance for your product application

Certificate awarded by EFY

Robotics (Delhi)

Duration: 10 days (60 hours); **Schedule:** Delhi: 9 Nov - 20 Nov; **Course Fee:** Rs 8000/-*

COURSE DESCRIPTION:

Science and engineering of robotics; electronic and mechanical hardware related to robotics; basics of PIC16F877A applications used in robotics. Project work: Construction of a robot based on PIC16F877A MCU TAKE HOME YOUR OWN ROBOT AFTER COMPLETION OF THE COURSE